

Steroid-Induced Cataract: New Perspectives from *In vitro* and Lens Culture Studies

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The prevailing view regarding the mechanism of steroid cataract formation holds that glucocorticoids are covalently bound to lens proteins resulting in destabilization of the protein structure allowing further modification (i.e. oxidation) leading to cataract. Alternative hypotheses (e.g. that cataracts result from glucocorticoid receptor mediated effects) have been difficult to test since protein binding does in fact occur for many cataractogenic steroids. A glucocorticoid lacking the typical glucocorticoid hydroxy group at C21 (fluorometholone, FML), other steroids which can bind to proteins but lack glucocorticoid activity, and a glucocorticoid antagonist (RU486) have been utilized to discriminate between these two hypotheses. Purified bovine β -crystallin incubated with three different ³H-steroids, dexamethasone (Dex), aldosterone or progesterone demonstrated that the C-21 hydroxyl group is not essential for steroid binding. Progesterone (with no C-21 OH) bound to the greatest extent. Pretreatment of the protein with aspirin to acetylate the free protein amino groups blocked this binding, demonstrating the probability of a Schiff base mechanism. Lens culture studies with the same three radiolabeled steroids demonstrated much the same result. Rat lenses cultured for 48 hr–11 days, demonstrated that loss of GSH is an early and significant effect of several glucocorticoids (Dex, prednisolone and FML) but is not seen with other non-glucocorticoid steroids. However, none of the steroids tested consistently produced lenticular opacification (i.e. cataracts) in this *in vitro* system, nor did they alter rubidium transport. We suggest that a mechanism other than covalent binding of steroids to lens proteins is responsible for glucocorticoid induced cataracts because: (1) non-glucocorticoids were demonstrated to bind lens proteins as well or better than the glucocorticoid Dex and (2) only glucocorticoids, and not other steroids, lowered lens reduced glutathione content which has been demonstrated to be associated with other forms of cataract.

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1. Introduction

The development of posterior subcapsular cataracts (PSC) following prolonged use of topical or systemic glucocorticoids has been the subject of a great many clinical reports over the last four decades (e.g. Black et al., 1960; Havre, 1965; Williamson et al., 1969; Urban and Cotlier, 1986; Kaye et al., 1993). Although a clear correlation between duration of treatment or dosage and incidence of opacities has not emerged (Kaye et al., 1993), it appears that the combination of these two variables as well as the route of administration (topical vs. systemic) play roles in development of steroid-induced cataracts (Urban and Cotlier, 1986). As a common consequence of prolonged ocular steroid treatment, it would be useful to better understand how steroids cause cataracts.

The mechanism of steroid-induced cataract is somewhat obscure. Some have suggested that steroids cause an inhibition of the cation pump and the resulting electrolyte/water imbalance is responsible for cataract formation (Mayman, Miller and Tijerina, 1979). Others believe steroid cataract to have an oxidative mechanism (Creighton et al., 1983; Nishigori et al., 1985; Costagliola et al., 1987, 1993). The

most prominent hypothesis involves the formation of Schiff bases between the steroid C-20 carbonyl group and ϵ -amino groups of crystallin lysine residues. A Heyns rearrangement involving the adjacent C-21 hydroxyl, results in a stable ketoimine product (Fig. 1). The covalently attached steroid is postulated to destabilize the protein's conformation, 'opening up' its structure and allowing the oxidation and cross-linking of the protein SH groups (Bucala et al., 1982, 1985a, 1985b; Manabe, Bucala and Cerami, 1984; Urban and Cotlier, 1986). Central to this hypothesis is the steroid structural requirement of a ketol functionality, i.e. a keto group available for Schiff base formation and an adjacent OH group which is free to participate in the subsequent Heyns rearrangement.

The Bucala hypothesis makes several tacit assumptions. First, only steroids with a hydroxyl group adjacent to the reactive carbonyl can form stable protein-steroid adducts and thus be cataractogenic (Although it is hard to imagine why steroids which are not irreversibly bound would not also alter conformation). Second, any steroid with this type of ketol functionality should prove cataractogenic. Third, the cataractogenic potential of a steroid is not dependent on its glucocorticoid activity. To properly test this hypothesis, each of these points should be addressed. The work described in this report evaluates these

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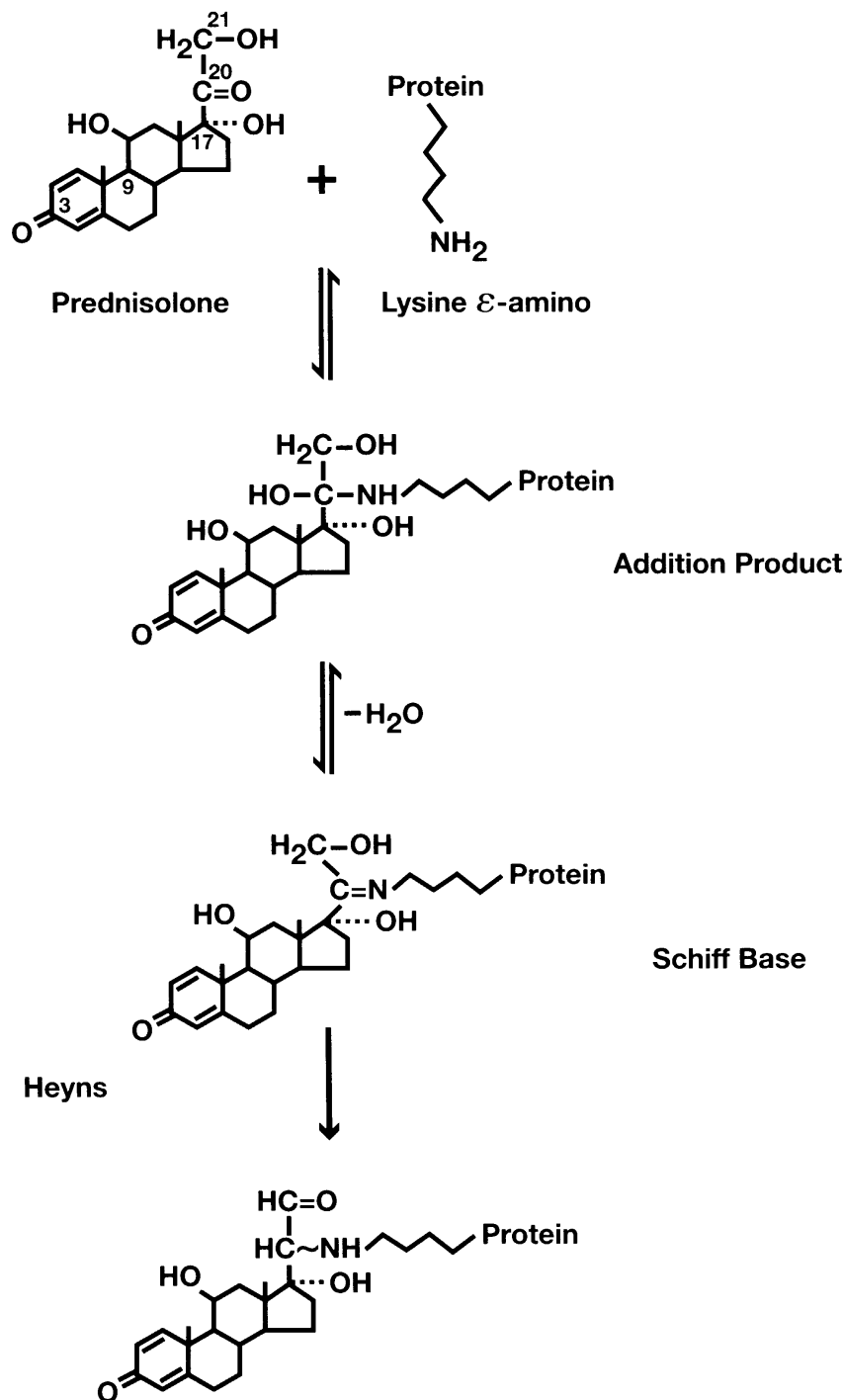


FIG. 1. Schiff base formation and subsequent Amadori product following Heyns rearrangement between a steroid (prednisolone) and a protein lysyl residue.

assumptions using both in vitro methods with purified bovine β -crystallin, and ex vivo rat lenses cultured in the presence of steroids.

2. Methods

Chemicals

Bovine β -crystallin, diethylenetriaminepentaacetic acid (DTPA), trichloroacetic acid (TCA), acetylsalicylate, ouabain, aldosterone, dexamethasone (Dex), progesterone, prednisolone and fluorometholone

(FML) were obtained from Sigma (St. Louis, MO, U.S.A.). 4-Pregnen-11 β ,17 α ,21-tetrol-3-one (Preg) was the product of Steraloids (Wilton, NH, U.S.A.). RU486 was obtained from Roussel-UCLAF. All other chemicals used were of analytical grade and were obtained from Sigma unless otherwise noted. Steroid structures are depicted in Fig. 2.

Radiochemicals

[1,2-³H]Aldosterone (48.0 Ci mmole⁻¹), [1,2,4-³H]Dex (38.0 Ci mmole⁻¹), [1,2,6,7,16,17-³H]Proges-

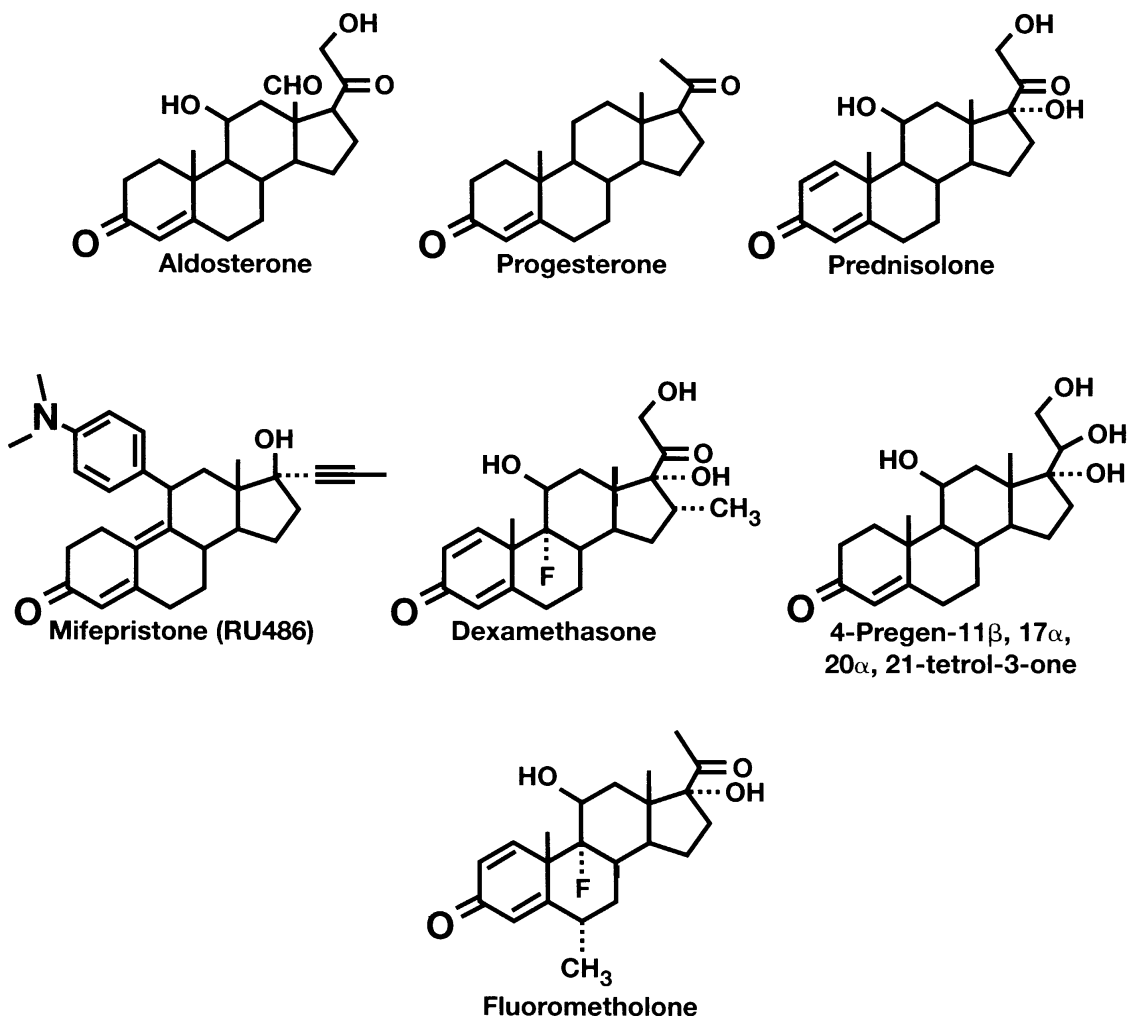


FIG. 2. Molecular structures of steroids used in these studies.

terone (103 Ci mmole⁻¹) and ⁸⁶Rubidium chloride (1 mCi ml⁻¹) were obtained from Amersham Life Science (Arlington Heights, IL, U.S.A.).

In Vitro Bovine β -Crystallin Incubations

Bovine β -crystallin was dissolved (1 mg ml⁻¹) in potassium phosphate buffer (0.1 M), 10 mM DTPA, pH 7), purged with nitrogen and dispensed into amber vials. Vials were randomly assigned to one of four treatment groups (3 per group), Control (no steroid, 15 μ l ethanol), ³H-Dex (15 μ l, 15 μ Ci, 200 nM), ³H-Aldosterone (15 μ l, 15 μ Ci, 158 nM), and ³H-Progesterone (15 μ l, 15 μ Ci, 73 nM). One week following addition of the steroid or ethanol, the β -crystallin was precipitated by the addition of 100% TCA (w/v) (10% final concentration), and centrifuged at low speed for 10 min. The pellet was resuspended and washed three additional times in 10% TCA. The protein-associated radioactivity measured is assumed to reflect covalent attachment of steroids as has been shown by others (Bucala et al., 1982).

To assess the nature of the binding between the steroids and the β -crystallin, the crystallin amino

groups were acetylated by pretreatment with aspirin (Rao, Lardis and Cotlier, 1985). Bovine β -crystallin was pretreated with or without acetylsalicylic acid (15 mM) at 37 °C for 24 hr prior to the steroid incubation. The β -crystallin was then incubated with the three steroids as described above. For this experiment, three methods were employed to remove the unbound steroid; TCA washing, or extensive dialysis with or without a ten-fold excess (10⁻⁶ M) of cold (unlabeled) steroid in the final dialysis buffer. Dpm for all three samples were then measured.

Ex Vivo Rat Lens Culture: Binding Studies

To more closely approximate in vivo conditions, rat lens culture was employed as previously described (Zigler and Hess, 1985; Dickerson, Lou and Gracy, 1995). Lenses were incubated with either 10⁻⁶ M ³H-Dex, 10⁻⁶ M ³H-progesterone, 10⁻⁶ M ³H-aldosterone (all at 1.0 μ Ci nmole⁻¹ in 0.2% ethanol) or 0.2% ethanol (no steroid). Incubation then proceeded for one week with media replacement every 48 hr. Following one week, lenses were harvested as follows. Lenses were removed from the media, rinsed in sterile

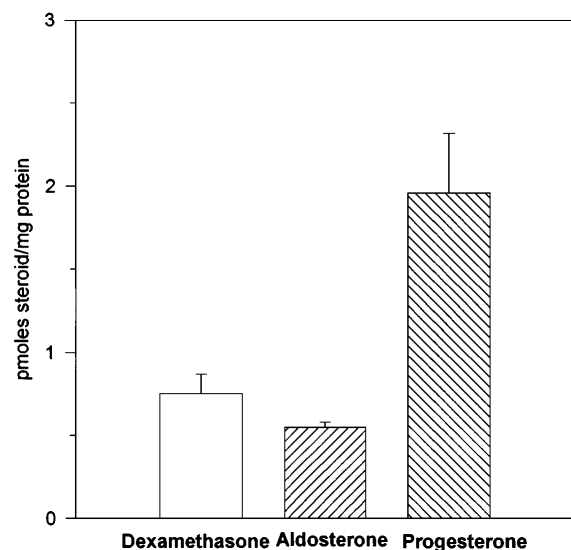


FIG. 3. Formation of β -crystallin/steroid adducts following one week incubation of the protein (2 ml, 1 mg ml⁻¹) with 15 μ Ci of dexamethasone (1.97×10^{-7} M), aldosterone (1.56×10^{-7} M), or progesterone (7.28×10^{-8} M). Data are mean $\pm$ one S.D. $n = 3$. Aldosterone and dexamethasone are both significantly different from progesterone ($P < 0.05$), but are not significantly different from each other (one-way ANOVA/SNK test).

saline, weighed and homogenized in 1.0 ml ice-cold 10% TCA. The TCA precipitates were washed three times in fresh TCA and resuspended in 8 M urea prior to adding to scintillation cocktail for dpm measurement.

An additional experiment was carried out to more completely explore the nature of progesterone binding in the cultured rat lenses. Three groups of rat lenses ($n = 5$ for each) were cultured with ³H-progesterone as outlined above. Following one week they were harvested and homogenized in either ice-cold 10% TCA or in 0.1 M potassium phosphate, 1 mM DTPA, pH 7. The TCA precipitates were then centrifuged, washed and assayed for radioactivity as described above while the phosphate buffer homogenates were extensively dialysed prior to measurement of dpm.

Lens Transparency Studies

Two studies were carried out in an effort to reproduce published experiments in which prednisolone at concentrations as low as 5×10^{-9} M could produce nuclear opacities in rat lenses cultured for 11 days (Manabe et al., 1984). Culture methods for these experiments were exactly as stated in Manabe et al. (1984). Lenses were cultured with prednisolone (5 and 50 nM, 0.01% ethanol), 4-pregnen-11 β ,17 α ,20 α ,21-tetrol-3-one (Preg) (a non-ketolic analog of prednisolone, 5 and 50 nM, 0.01% ethanol) (Fig. 2) or in normal media with 0.01% ethanol. Just prior to harvesting, the lenses were graded for clarity by the criteria in Table I. After assessing lens clarity, lenses were assayed for glutathione (GSH) and protein sulfhydryl groups (PSH).

Sulphydryl Assays

PSH and GSH assays were based on Ellman's methodology (Ellman, 1959) with some modification (Lou et al., 1988).

Polyacrylamide Gel Electrophoresis (PAGE)

Aliquots of Tris buffer lens homogenates were used for SDS-polyacrylamide gel electrophoretic analysis under reduced and non-reduced conditions (Laemmli, 1970). Protein bands were stained with Coomassie brilliant blue (R-250) for visualization.

Protein-Thiol Mixed Disulfides

Lenses cultured for one week in either Dex (10^{-6} M) or normal media were assayed for GSH as described above and also for the presence of protein-GSH (PSSG) or protein-cysteine (PSSC) mixed disulfides via methodology thoroughly described elsewhere (Lou and Dickerson, 1992; Dickerson and Lou, 1993).

Glucocorticoid Blockade

The GSH changes associated with incubation in Dex containing media were studied by attempting to block glucocorticoid function through the addition of RU486, a potent glucocorticoid antagonist (Fig. 2), to the incubation media. Lenses were cultured in the presence of 2.5×10^{-5} M concentrations of Dex, progesterone, RU486, Dex + RU486, or control media. Lenses were harvested after 48 hr for glutathione determinations.

Statistical Analysis

Statistical analysis was performed with Sigma Stat (Jandel Scientific). Parametric tests were employed (ANOVA for multi-group comparison; *t*-test for comparison of two groups) unless parametric assumptions regarding normality of data and equality of variances were violated. In such cases the non-parametric counterparts were used. Significant ANOVA results were further analysed with the SNK test for multiple comparisons (parametric) or Dunn's test (non-parametric).

TABLE I
Lens clarity grading scale

Grade	Morphology
0	Clear lens
1	Haziness; small peripheral opacities
2	Larger, more contiguous peripheral opacities
3	Haziness, central nuclear shell begins to form
4	Central shell opaque
5	Shell fully thick and white, grows to cortex.

3. Results

In Vitro Steroid Binding

The results of the in vitro β -crystallin binding experiment are shown in Fig. 3. Aldosterone and Dex both have C-20 keto/C-21 hydroxy arrangements and both bound to β -crystallin to a similar degree ($0.75 \text{ pmol mg}^{-1}$ protein for Dex, $0.55 \text{ pmol mg}^{-1}$ for aldosterone). The slightly better binding for Dex may be due in part to the greater steric hindrance associated with the C-11/C-18 hemiacetal in aldosterone. In contrast, progesterone does not have a hydroxyl group adjacent to a carbonyl anywhere on its structure yet was bound 2.6 X the value for Dex and 3.6 X that for aldosterone. This in spite of a concentration that was less than half that of Dex or aldosterone (due to differing specific activities).

Pretreatment of β -crystallin with aspirin (which acetylates amino groups) (Crompton, Rixon and Harding, 1985) blocked 98.4% of the progesterone binding (data not shown), implicating Schiff base formation in the progesterone binding observed when aspirin was not present. Dialysis of untreated β -

crystallin with a 'chase' of cold progesterone reduced the amount of bound labeled progesterone to nearly the level of the aspirin treated protein (data not shown). This demonstrates the reversible nature of the reaction and that an irreversible Amadori product has not formed. High levels of binding were only observed in samples which had been subjected to TCA washes to isolate the protein. Aspirin pre-treatment did not decrease further the already very low level of binding by either aldosterone or Dex.

Steroid Binding in Rat Lens Culture

In agreement with the in vitro data, progesterone was bound to the lens proteins in the greatest abundance ($2.95 \text{ pmol mg}^{-1}$ protein, expt. 1; $2.25 \text{ pmol mg}^{-1}$, expt. 2) followed by Dex ($0.36 \text{ pmol mg}^{-1}$ protein) and aldosterone ($0.02 \text{ pmol mg}^{-1}$ protein) (Fig. 4). In fact, though the level of progesterone bound in the lens is similar to the in vitro data, Dex binding in the lens was about one half as much as in vitro and aldosterone binding was nearly 30-fold less. When dialysis was used to remove the unbound progesterone, only 0.4 pmol mg^{-1} progesterone remained associated with the protein. Interestingly, the ratio of bound progesterone after dialysis to bound progesterone after TCA treatment was substantially larger in the lens culture experiments (17.8%) than for the in vitro studies (2.3%). This could result from the unique environment inside the lens where the high levels of free GSH could reduce the double bond of the Schiff base yielding an irreversibly bound adduct (Jocelyn, 1972).

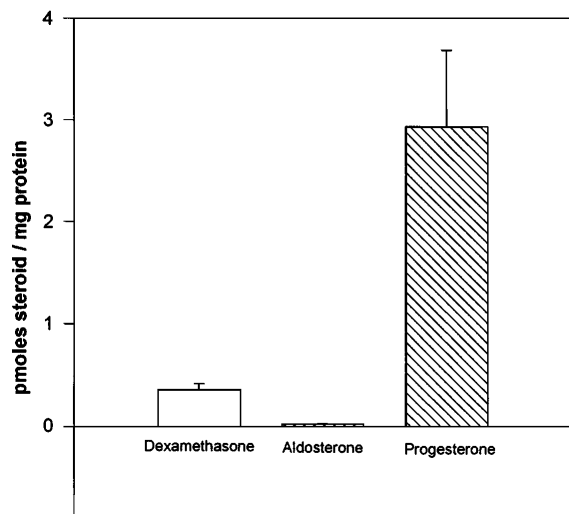


FIG. 4. Steroid binding to lens proteins during one week lens culture. Culture details are given in Methods. Steroids were 10^{-6} M , $1 \mu\text{Ci nmole}^{-1}$. Data are mean $\pm$ one s.d. $n = 4$. Each of the three groups is significantly different from the other two ($P < 0.05$) (one-way ANOVA/SNK test).

Lens Clarity

Lenses cultured for 11 days in 5 or 50 nM prednisolone or in the same concentrations of its non-keto C20 analog were found to be no different from controls in terms of lens clarity scores. Note that all groups had at least one lens that was judged to be perfectly clear. While the two prednisolone groups had slightly higher average scores than the control group (1.9 and 2.0 vs. 1.2), the 5 nM non-keto analog (Preg) group was actually highest at 2.4. None of these differences were found to be significant (Kruskal-Wallis test, $P = 0.75$). This experiment was repeated with essentially the same results.

Lens GSH and PSH

Lens PSH levels were measured for the lenses from the 11 day lens clarity studies. The mean $\pm$ s.d. for each group are given in Table II. These values ranged between $24.6 \pm 0.7 \mu\text{moles g}^{-1}$ for the controls ($n = 6$) and $20.4 \pm 1.2 \mu\text{moles g}^{-1}$ for the low concentration of prednisolone (5 nM, $n = 5$). This was the only group significantly different ($P < 0.05$) from the controls. Note that the high prednisolone concentration (50 nM,

TABLE II
Protein Sulfhydryl Groups (PSH)

Treatment group	5 nM	50 nM
Control	24.6 ± 0.7	24.6 ± 0.7
Prednisolone	$20.4 \pm 1.2^*$	22.3 ± 2.1
4-Pregnen-11 β , 17 α , 20 α , 21-tetrol-3-one	23.1 ± 2.6	23.2 ± 0.5

Data are $\mu\text{mole g}^{-1}$ lens weight, mean $\pm$ standard deviation (s.d.), $n = 6$.

* $P < 0.05$, one-way ANOVA/SNK multiple comparison test.

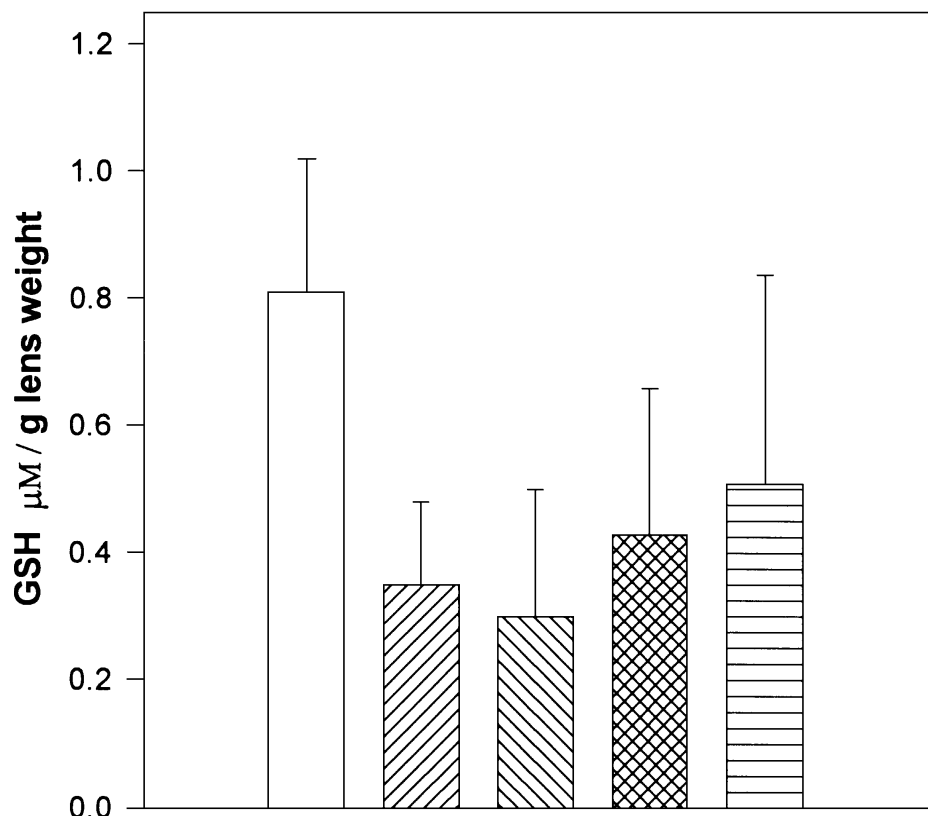


FIG. 5. Lens GSH concentration from rat lenses cultured for 11 days with prednisolone or 4-pregnen-11b, 17a, 20a, 21-tetrol-3-one. All steroid treated groups have significantly lower GSH levels than control ($P < 0.05$), one-way ANOVA/SNK test. Data are mean $\pm$ one s.d. $n = 6$. $\square$, control; diagonal lines , 5×10^{-9} M prednisolone; cross-hatch , 5×10^{-8} M prednisolone; horizontal lines , 5×10^{-9} M 4-pregnen-11b, 17a, 20a, 21-tetrol-3-one; vertical lines , 5×10^{-8} M 4-pregnen-11b, 17a, 20a, 21-tetrol-3-one.

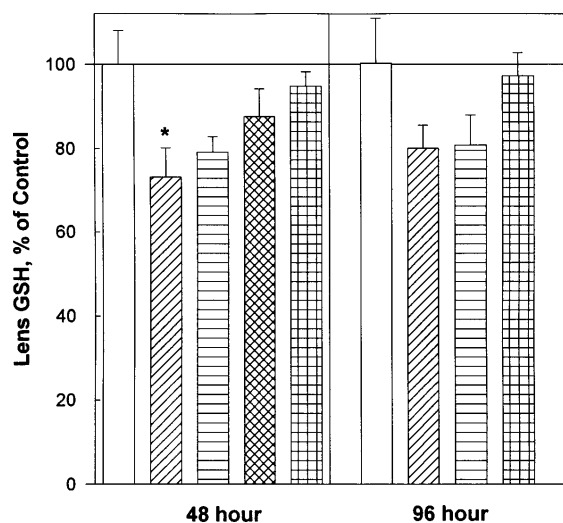


FIG. 6. Lens GSH in 48 and 96 hr rat lens culture experiments. $\square$, control; diagonal lines , Dex (10^{-6} M); horizontal lines , FML (10^{-6} M); cross-hatch , Dex (2.5×10^{-5} M) + RU486 (2.5×10^{-5} M); vertical lines , Progesterone (10^{-6} M). Forty-eight hour Dex treated group significantly different from control. ($P < 0.05$), one-way ANOVA/SNK test.

$n = 6$) group was not different from the controls ($22.3 \pm 2.1 \mu\text{moles g}^{-1}$ vs. 24.6 ± 0.7). Lens GSH levels were found to be uniformly low (compared to in vivo levels) for all groups ($0.35\text{--}0.81 \mu\text{mole g}^{-1}$) (Fig. 5). Although the two prednisolone groups were the lowest

in lens GSH, all of the steroid treated groups were found to be significantly lower in lens GSH than the controls ($P < 0.05$, SNK test).

The finding that the prednisolone treated groups were lowest in GSH suggested an effect on lens redox status mediated by glucocorticoids. A follow-up experiment (Fig. 6) examined lenses cultured for shorter intervals (48 and 96 hr) since the 11 day culture period represented a significant challenge to the maintenance of GSH levels in and of itself (Dickerson et al., 1995). Other glucocorticoids were used to determine if GSH depletion was a phenomenon common to all glucocorticoids. Lenses cultured for 48 hr in either Dex (10^{-6} M) or FML (10^{-6} M) had GSH levels only 73% or 79% of control ($2.36 \mu\text{mole g}^{-1}$), respectively, but had declined no further at 96 hr (Dex, 80% and FML, 81% of control; control = $2.39 \mu\text{mole g}^{-1}$). A statistically significant difference could only be demonstrated between the control and Dex groups ($P < 0.05$, one-way ANOVA and SNK multiple comparison test).

Lenses cultured for 48 hr with 2.5×10^{-5} M Dex had only 74% of the control GSH level, not significantly different from the previous result in spite of a 25-fold greater Dex concentration (data not shown). However, addition of 2.5×10^{-5} M RU486, a glucocorticoid antagonist, to the media along with the Dex prevented a substantial portion of the GSH loss, lenses had 90%

TABLE III

Mixed disulfide levels from rat lenses cultured for 96 hr in Dex or control media

Treatment group	PSSC	PSSG
Control	390 ± 100	27.2 ± 9.1
Dexamethasone	510 ± 100	33.3 ± 16.8

Data are nmole g⁻¹ lens weight, mean ± s.d., *n* = 5. Mixed disulfides are measured as either glutathione sulfonic acid or cysteic acid (see Methods).

Differences between control and Dex groups are not statistically significant, *P* > 0.05, one-way ANOVA.

of the control level (not significantly different from the control group). Progesterone, a steroid devoid of glucocorticoid activity, at equimolar concentrations did not affect lens GSH (95% and 96% of control at 48 and 96 hr, respectively) (Fig. 6).

Protein-Protein Crosslinking

Comparison of protein banding patterns on polyacrylamide electrophoretic gels for lenses from the lens clarity studies revealed no striking differences between controls, Dex, or Preg treated lenses in either reduced or non-reduced samples (data not shown).

Protein-Bound Thiols

Analysis of lenses cultured for 1 week with 10⁻⁶ M Dex showed a 22.4% increase in PSSG over the controls (Table III). Interestingly, protein-cysteine mixed disulfide (PSSC) was also elevated (30.8%). Neither of these increases was found to be statistically significant at this time point. The PSSG formed accounts for only about 1.3% of the total GSH loss in the Dex treated lenses.

Lens Transport Function

Membrane integrity and transport function of rat lenses cultured 48 hr in 10⁻⁶ M Dex or 10⁻⁶ M FML was assessed through their ability to concentrate ⁸⁶Rb from the media. No significant differences were detected between control and either glucocorticoid treated group (data not shown).

4. Discussion

The Bucala et al. (1982, 1985a) hypothesis for steroid cataract is similar in concept to the glycation mechanism hypothesis for senile cataract as originally formulated by Stevens et al. (1978) and others (Monnier, Stevens and Cerami, 1979). In essence both contend that molecules with a keto or aldo group adjacent to a carbon bearing hydroxyl may form Schiff bases with protein amino groups and that these

adducts (steroid or glyco moieties) alter the conformation of the protein such that previously inaccessible sulfhydryls are now liable to oxidation and participation in protein-protein crosslinks. When this occurs to a sufficient extent (according to the hypothesis), cataracts develop.

The modification of lens proteins by steroids has been documented (Manabe et al., 1984; Bucala et al., 1985a, 1985b). It should be noted however, that this modification occurs to a very limited extent. Data from the present study showed that after one week of incubation with purified β-crystallin, approximately one β-crystallin subunit in 51000 was modified by Dex. Even human lenses from patients subjected to long-term steroid therapy show very low levels of steroid-protein adducts. Manabe et al. (1984) reported modification levels of 5–67 pmol prednisolone mg⁻¹ protein in human cataracts (approx. 1 modification per 600–8000 protein molecules). It is thus not clear what role this phenomenon plays in the cataractogenic process.

Three corollaries emerge from the steroid-adduct hypothesis of cataractogenesis: All steroids with C20 carboxyl/C21 hydroxyl arrangements should bind to lens proteins and result in cataracts. Cataractogenic potential of a steroid should be unrelated to its hormonal activity (or lack thereof). Steroids without the requisite C20/C21 structure should not be cataractogenic.

The first of these, that all steroids with the appropriate structure at C20/C21 will bind lenticular proteins and therefore cause cataracts is difficult to prove or to discount based on the published clinical literature. All accounts of steroid cataract in the literature involve glucocorticoids. The fact that most commonly prescribed glucocorticoids do have a structure promoting Schiff base formation makes the discrimination of effects due to protein modification and those due to glucocorticoid function difficult (but see FML clinical data discussed below). There are non-glucocorticoids with the same C20/C21 structure. However, they are either not commonly used as ophthalmic drugs, or in cases of disease where they are overproduced by the body, as in aldosteronism, they are very efficiently removed from circulation by the liver (Horton, 1973) and never reach potentially cataractogenic levels.

In the present study progesterone, which has a C20 carbonyl group but without the prescribed C21 hydroxyl, bound lens proteins eight-fold higher than Dex and nearly 150-fold higher than aldosterone in lens culture experiments. It is likely that differing lipophilicities affect the diffusion of these compounds into the lens interior. This characteristic, as measured by octanol logP values (Hansch et al., 1995), (progesterone, 3.8 > Dex, 2.0 > aldosterone, 1.1), allows progesterone to more readily enter and potentially diffuse through the network of lipid bilayers in the lens. Aldosterone, a much more polar molecule,

was probably limited to extracellular spaces of the outer cortex, similar to other molecules with logP values approximately 1 or less (Tang-Liu, Richman and Liu, 1992).

The facility with which progesterone binds lens proteins *in vitro* or in cultured lenses is not physiologically relevant. The human lens probably never encounters significant levels of progesterone. Although progesterone may be synthesized at relatively high levels in ovulating females ($2\text{--}8\text{ mg } 24\text{ hr}^{-1}$) it is rapidly and efficiently metabolized by the liver to pregnanediol (Berczeller, Young and Kuperman, 1964). Pregnanediol has no carbonyl group and thus cannot form a Schiff base with a protein.

The fact that progesterone has no inherent activity as a glucocorticoid yet binds efficiently with lens proteins makes it a useful tool to discriminate between effects mediated through the glucocorticoid receptor and those resulting from Schiff base formation with proteins. If pre-cataractous biochemical changes in rat lenses are noted following culture with progesterone, one could conclude that protein binding may be an important initial component of steroid cataractogenesis. On the other hand, if no such changes are observed, but are found in Dex treated lenses, the conclusion must be that something unique about Dex when contrasted with progesterone, is responsible. GSH depletion is an early hallmark of cataractogenesis in virtually every type of cataract known (Sippel, 1966; Harding, 1970; Kuck, 1970; Anderson, Wright and Spector, 1979; Lou et al., 1988, 1990; Giblin et al., 1995). Although GSH depletion has not been directly linked with the formation of steroid-induced PSC, several lines of evidence support such a role. (1) Loss of lens GSH has also been reported from rabbit eyes treated with corticosteroids for 40 days (Costagliola et al., 1987). Betamethasone ($11.8\text{ }\mu\text{g}^{-1}\text{ eye}^{-1}\text{ day}^{-1}$) and FML ($11.3\text{ }\mu\text{g}^{-1}\text{ eye}^{-1}\text{ day}^{-1}$) induced 68% and 51% drops in GSH respectively. The GSH concentration in the aqueous humor was not changed indicating that the drop in lenticular GSH was not secondary to an exterior decrease. (2) Creighton et al. (1983) found that rat lenses cultured with solumedrol (methylprednisolone-21-succinate) developed cortical opacities which could be significantly reduced if vitamin E was included in the culture media. This implies some type of oxidative mechanism was involved. (3) Korte and Hockwin (1978) demonstrated that prednisolone inhibits glucose-6-phosphate dehydrogenase in calf lens. Since this enzyme catalyses a reaction which is a major source of the cellular NADPH supply, inhibition could have serious consequences for any reaction or process requiring NADPH. Recycling of oxidized glutathione (GSSG) back to GSH via glutathione reductase is one such process. (4) In chick embryos treated with hydrocortisone hemisuccinate, lenticular GSH losses were reported which could be ameliorated by treatment with ascorbic acid (Nishigori et al., 1985). Decline in endogenous GSH

and ascorbic acid levels paralleled the appearance of lenticular opacity. Recovery of transparency was concomitant with normalization of these two parameters. (5) In the present study note that progesterone had no significant effect on lens GSH levels at either 48 or 96 hr while Dex treated lenses over the same time period had lost 20–30% (Fig. 6). The presence of RU-486, a potent glucocorticoid antagonist prevented much of this loss. These studies strongly suggest that it is glucocorticoid activity mediated through the glucocorticoid receptor that is responsible for the loss of GSH from these lenses, and is perhaps responsible for cataractogenesis.

FML, a potent glucocorticoid, has a unique structure compared to other glucocorticoids. The C17 side-chain of FML lacks the C21 hydroxyl group common to most other glucocorticoids. This structure should prevent irreversible binding to protein amino groups and thus be non-cataractogenic (Manabe et al., 1984). In our studies, FML equalled Dex in ability to deplete lens GSH. Similar results have been reported by others in rabbit lenses (Costagliola et al., 1987; 1993). FML has been implicated as the causative agent in at least five clinically documented cases of posterior subcapsular cataract (Costagliola et al., 1989; Muñoz-Negrete and Arenas-Archila, 1989) and possibly a sixth (Parker and Binder, 1979). The occurrence of PSC with FML further supports the position that it is the glucocorticoid nature of the steroid that is important, not the protein binding potential of its side chain.

The studies reported here have captured changes that commonly occur in the early stages of several types of cataract. The lens clarity studies, in direct contrast to those of Manabe et al. (1984), noted no clarity differences between glucocorticoid treated and control lenses. The covalent adduct hypothesis speculates that an early step in the steroid cataractogenic process is an 'opening up' of the protein structure allowing oxidation of sulfhydryls. This would then account for GSH depletion and should be accompanied by protein-protein crosslinking. No increase in crystallin dimers or multimers, exclusive to the glucocorticoid treated group, consistent with such changes, was seen at these early stages. Rather, the loss of free GSH was the first obvious change. It is more likely that it is this GSH change which precedes and potentiates the protein changes that occur as the cataract develops (Lou, Dickerson and Garadi, 1990). Although the extent of GSH loss seen here may represent a relatively minor insult for the lens as a whole, for the critical areas involved in cell division and differentiation it could be much more significant.

As alluded to above, the region posterior to the bow of the lens is of critical importance in the normal development of fiber cells. Other types of insult have been shown to manifest their effect in this region resulting in PSC (Zigler and Hess, 1985). The presence of glucocorticoid receptors in the epithelium immediately to the anterior, as shown for bovine lens (Wenk

et al., 1982), gives further credence to the hypothesis that glucocorticoids are producing their effects via activation of this receptor. Further experiments are being undertaken to clarify this point.

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